



PAPER ID-311604

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Subject Code: BMI102

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B.TECH
(SEM I) THEORY EXAMINATION 2024-25
MATHEMATICAL FOUNDATION

TIME: 3 HRS**M.MARKS: 70****Note:** Attempt all Sections. In case of any missing data; choose suitably.**SECTION A****1. Attempt all questions in brief.****2 x 07 = 14**

Q no.	Question	CO	Level
a.	What is a row-reduced echelon form of a matrix?	1	K2
b.	Define a vector in 2D and 3D space with an example.	2	K2
c.	What is the Chain Rule in differentiation? Give an example.	3	K3
d.	State the properties of Beta and Gamma functions.	4	K3
e.	What is the general solution of a linear differential equation?	4	K3
f.	What is the Laplace Transform of a constant function?	5	K1
g.	What is the Laplace Transform of a derivative?	5	K2

SECTION B**2. Attempt any three of the following:****7 x 3 = 21**

Q no.	Question	CO	Level
a.	Prove that the determinant of a matrix remains unchanged under a row swap operation, but changes sign. Verify with an example.	1	K3
b.	Discuss the geometric significance of eigenvectors and eigenvalues in transformations.	2	K2
c.	Using Leibnitz's Theorem, find the fourth derivative of $y = e^x \sin x$.	3	K3
d.	Evaluate using Gamma function properties. $\int_0^{\infty} e^{-x} x^2 dx$	4	K4
e.	Derive the Laplace Transform of t^n .	5	K2

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

Q no.	Question	CO	Level
a.	Verify whether the matrix is singular or non-singular. $D = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$	1	K2
b.	Explain the process of finding the inverse of a matrix using elementary row operations.	1	K3



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TIME: 3 HRS**M.MARKS: 70****4. Attempt any one part of the following:****07 x 1 = 07**

Q no.	Question	CO	Level
a.	Verify the Cayley-Hamilton theorem for $D = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	2	K2
b.	Solve the system of equations using matrix inverse method: $x + 2y + 3z = 9$ $2x + 5y + 8z = 19$ $3x + 6y + 9z = 24$	2	K2

5. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Verify Euler's theorem for $f(x, y) = 2x^2 + 3y^2$	3	K3
b.	Compute the Jacobian determinant for the transformation: $u = x^2 + y^2, v = x^2 - y^2, w = xy$	3	K4

6. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Find the particular integral of $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x.$	4	K3
b.	Explain the method to solve linear differential equations with constant coefficients.	4	K3

7. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Explain the properties of Adjacency Matrices of graphs.	5	K2
b.	Prove that a tree with n vertices has (n-1) edges.	5	K2